

Security in Cloud Computing

Module 38

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Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agriculture University, Ludhiana

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Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture we will study about Security in cloud computing first of all please see the contents which I shall cover after covering brief introduction about Security in cloud computing I shall take up network security in cloud and last I shall take up platform related security so now let us start security is an important aspect to be considered in the cloud computing environment cloud computing security refers to the C of procedures processes and standards designed to provide information security Assurance in an cloud computing environment cloud computing security addresses both physical and logical security issues across all the different

service models of software platform and infrastructure it also addresses how these services are delivered public private or hybrid delivery model Cloud security encompasses a broad broad range of security constraints from an end user and Cloud provider's perspective where the end user will primarily will be concerned with the provider security policy how and when their data is stored and who has access to that data subsequent sections talk about data security virtualization security security issues in software as a service infrastructure as a service and platform as a service model Etc security aspects cloud computing places business data into the hands of an outside provider and makes Regulatory Compliance inherently riskier and more complex than

it is when systems are maintained in house loss of direct oversight means that the client company must verify that the service provider is working to ensure that data security and integrity are IR cled the following are the current security related research areas in cloud computing reliable distributed applications based on the internet such as the e-commerce system rely heavily on the trust path among involved parties the skyrocketing demand for a new generation of cloud-based consumer and business applications is driving the need for the next generation of data centers that must be massively scalable efficient agile reliable and secure in order to scale cloud services reliable to millions of service developers and

billions of end users the Next Generation cloud computing and data center infrastructure will have to follow an evolution similar to the one that led to the creation of scalable telecommunication networks in the future network-based cloud service providers will leverage virtualization Technologies to be able to allocate just the right levels of virtualized compute Network and storage resources to individual applications based on realtime business demand while also providing full service level Assurance of availability performance and security at a reasonable cost data security due to huge infrastructure cost organizations are slowly switching to Cloud technology data are stored in the cloud service provider infrastructure as

data do not reside in organization territory many complex challenges arise some of the complex data security challenges in Cloud include the following cloud service models with multiple tenants sharing the same infrastructure data mobility and legal issues relative to such government rules as European Union data privacy directive lack of Standards about how cloud service providers securely recycle disk space and erase existing data laws of visibility to key security and operational intelligence that no longer is available to feed Enterprise it security intelligence and risk management a new type of Insider who does not even work for your company but may have control and visibility into your data data center security data are stored in outside territory of the user in a location

called as data center which is unknown to the user as the location of the data center is unknown to the user it becomes a virtual data center the backbone of this virtual data center is virtual infrastructure or the virtual machine however virtual platforms are dependent on many other often forgotten components of both the physical and virtual data centers there are typically seven areas of concern that accompany any major virtual platform implementation or migration often these issues are not seen during staging and testing and only appear when the virtual machines take on the same amount of load as physical machines the critical points represent two Corner Stones of the data center Network and storage lack of performance and availability virtualization moves many

input output tasks tuned for Hardware to software via the hypervisor the virtualization translation layer is responsible for translating the optimized code for the software chip to the physical chip or CPU running on the underlying Hardware lack of application awareness one of the limitation of hypervisor and kernel based virtualization Solutions is that they only virtualize the operating system OS virtualization does not virtualize nor is it even aware of applications that are running on the OS even the same applications do not realize that they are using virtual Hardware on top of a hypervisor overflowing storage Network although converting physical machines to Virtual machines is an asset

for building Dynamic data centers hard drives become extremely large flat file virtual dis images consequently file storage becomes unmanageable congested storage Network due to the portable nature of os virtualization there can be Dramatical increase in data traversing the storage Network for the same reason that virtual machine disk files can overrun physical

storage once these images are made portable it becomes trivial to move these virtual machine images across the network from one host to another or from one storage array to another network security cloud is based on networking of many things together like the network of infrastructure while the network is the backbone of the cloud many challenges are encountered in this network some of the challenges in the existing Cloud

networks are discussed in the following application performance Cloud tenant should be able to specify bandwidth requirements for applications hosted in the cloud ensuring similar performance to the on premise deployments many tiered applications requires some guaranteed bandwidth between server instances to satisfy user transactions within an acceptable time frame and meet predefined service level agreements insufficient bandwidth between these servers will impose significant latency on user interactions flexible deployment of Appliance flexible deployment of appliances Enterprises deploy a wide variety of security appliances in their data centers such as deep packet inspection or intrusion detection systems and firewalls to protect their applications from attacks these are

often employed alongside other appliances that perform load balancing caching and application acceleration when deployed in the cloud an Enterprise application should continue to be able to flexibly exploit the functionality of these appliances policy enforcement complexities traffic isolation and access control to end users are among the multiple forwarding policies that should be enforced these policies directly impact the configuration of each router and switch changing requirements different protocols open shortest path first link aggregation group Virtual Router redundancy protocol vrrp and different flavors of L2 spanning tree protocols along with vendor specific protocols make it extremely challenging to build operate and interconnect a cloud Network at

scale application rewriting applications should run out of the box as much as possible in particular for IP addresses and Network dependent failover mechanisms applications may need to be Rewritten or reconfigured before deployment in the cloud to address several Network related limitations two key issues are first lack of an broadcast domain abstraction in the cloud Network and second Cloud assigned IP addresses for virtual servers location dependency Network appliances and servers are typically tied to a statically configured physical Network which implicitly creates a location dependent constraint for instance the IP address of a server is typically determined based on the vlan or the subnet to which it belongs Vlan and subnets are based on the physical switch

Port configuration therefore a virtual machine cannot be easily and smoothly migrated across the network constrained virtual machine migration decreases the level of resource utilization and flexibility besides physical mapping of Vlan or subnet space to the physical ports of a switch often leads to a fragmented IP address pool platform related security cloud service providers offer services in various service models like SaaS PaaS and IaaS where the user is offered varied Services based on his or her requirements every service

model offer brings with it many security related challenges like secured Network locality of resources accessing secure data data privacy and backup policy security issues in cloud service models cloud computing uses Three Delivery models

such as IaaS, PaaS and SaaS through which different types of Computing Services are provided to the end user these three delivery models provide infrastructure resources application platform and software as services to the cloud customer these service models play a different level of security requirements in the cloud environment IaaS is the basis of all cloud services with PaaS built upon it and SaaS in turn built upon it just as capabilities are inherited so are the information security issues and risks there are significant trade-offs to each model in terms of integrated features complexity versus extensibility and security if the cloud service provider takes care of only the security at the lower part of the security architecture

the consumers become more responsible for implementing and managing their security capabilities SaaS is a software deployed model in which applications are remotely hosted by the application or service provider and made available to customers on demand over the Internet the SaaS model offers the customers with significant benefits such as improved operational efficiency and reduced costs SaaS is rapidly emerging as the dominant delivery model for meeting the needs of Enterprise IT services PaaS is one layer above IaaS on the stack and abstracts everything up to OS middleware Etc it offers developers a service that provides a complete software development life cycle management from planning to design to

building applications to deployment to testing to maintenance everything else is abstract away from the view of the developers software as a service security issues in a traditional on premise application deployment model the sensitive data of each Enterprise continue to reside within the Enterprise boundary and are subject to its physical logical and personal security and access control policies SaaS applications are accessed through the web and so web browser security is very much important information security officers will need to consider various methods of securing SaaS applications web services security extensible markup language encryption SSL and available options used in enforcing data protection transmitted over the Internet this involves the use

of strong encryption techniques for data security and Fine grained authorization to control access to data the pain points of concern in SaaS are as follows network security in 16 minutes, 1 second in a SaaS deployment model sensitive data flow over the network needs to be secured in order to prevent leakage of sensitive information this involves the use of strong Network traffic encryption techniques such as the SSL and TLS for security resource locality in an SaaS model of a cloud environment the end users use the services provided by the cloud providers without knowing exact where the resources for such services are located due to the compliance and data privacy rules in various countries locality of data is of utmost importance in much

Enterprise architecture Cloud standards to achieve interoperability among clouds and to increase their stability and security Cloud standards are needed across organizations for example the current storage Services by a cloud provider may be incompatible with those for other providers in order to keep their customers Cloud providers May introduce so-called sticky services that create difficulty for the users if they want to migrate from one provider to the other data Access Data access issue is mainly related to security policies provide to the users while accessing the data the organizations will have their own security policies based on which each employee can have access to a particular set of data the security

policies May entitle some considerations wherein some of the employees are not given access to a certain amount of data these security policies must be adhered by the cloud to avoid intrusion of data by unauthorized users the saas model must be flexible enough to incorporate these specific policies put forward by the organization the model must also be able to provide organizational boundary within the cloud because multiple organizations will be deploying their business processes within a single Cloud environment data breaches since data from various users and business organizations lie together in a cloud environment breaching into the cloud environment will potentially attack the data of all the users thus the cloud becomes a high value Target

backup the saas vendor needs to ensure that all sensitive Enterprise data are regularly backed up to facilitate quick recovery in case of disasters also the use of strong encryption schemes to protect the backup data is recommended to prevent accidental leakage of sensitive information in the case of cloud vendors such as Amazon the data at RIS in est3 are not encrypted by default the users need to separately encrypt their data and backups so that it cannot be accessed or tampered with by unauthorized parties platform as a service security issues paas Pro provides a ready to use platform including OS that runs on vendor provided infrastructure as the infrastructure is of the cloud service provider various security challenges of the focused

architecture are caused mainly by the spread of the user objects over the hosts of the cloud stringently allowing excess of objects to the resources and defending the objects against malicious or corrupt providers reasonably reduce possible risks network access and service measurement bring together concerns about secure Communications and excess control well-known practices object scale enforcement of authorization and undeniable traceability methods May alleviate the concerns apart from the aforementioned problems user privacy must be protected in a public share Cloud therefore proper most solutions must be privacy aware service continuity is another concern for many Enterprises that consider Cloud adoption accordingly fall tolerant reliable

systems are required infrastructure as a service security issues cloud computing makes a lot of promises in the areas of increased flexibility and Agility potential cost savings and competition advantages for developers so that they can stand up and infrastructure quickly

and efficiently to enable them to develop the software to drive business success there are a lot of problems that cloud especially private Cloud solves but it is not that much good is solving problems related to security however in a private Cloud environment some of the traditional problems faced are as follows hypervisor security in private Cloud most or all of the services will run in a virtualized environment and the security model used by the hypervisor cannot be taken for granted a need to

evaluate the security models and the development of hypervisors becomes necessary multi-tenancy although all the tenants in the multi-tenancy environment will be from the same company not all tenants may be comfortable sharing infrastructure with other users within the same company identity management and access control in our traditional data center we were comfortable with the small handful of authentication repositories we had to work with active directory being one of the most popular but with private Cloud handling authentication and authorization for the cloud infrastructure handling denance and handling delegation of administration of various aspects of the cloud fabric are the major task to be addressed network security in private Cloud we are likely

to have many components of a service communicate with each other over virtual network channels only accessing the traffic employing some powerful access controls for physical networks and control quality of service which is a key issue in the availability aspect of the confidentiality integrity and availability security model are major concerns combining the three types of clouds public Cloud private cloud and hybrid Cloud together with the three Service delivery models we get a complete picture of a cloud computing environment interl by connectivity devices coupled with information security components virtualized physical sources virtualized infrastructure virtualized middleware platforms and business related applications are being provided

as Computing Services in the cloud cloud providers and Cloud consumers must be able to maintain and establish Computing security at all levels of interfaces in the cloud computing architecture so friends this was all about Security in cloud computing hope the concepts explained in this lecture were understandable and helpful hope to see you in next lecture till then goodbye thank you